



基于仿生鲫鱼吸盘的 导管架裂缝检测爬壁机器人

Bionic Remora-based Wall-climbing Robot
for Crack Detection in Conduit Racks

第十二届全国海洋航行器设计与制作大赛
全国一等奖

项目简介

项目基于柔性仿生鲫鱼盘及三足轮式结构，实现在导管架复杂表面持续吸附行走的功能，在攀爬过程中清除表面附着物，同时通过超声波探头不断检测导管架内部情况，实现吸附检测一体化，提高检测效率。项目基于仿生鲫鱼盘和三足轮式的吸附爬行结构给予了ROV更好的安全性和适应性，可以有效地帮助ROV在导管架表面进行稳定工作，提高裂缝检测结果的准确性，为水下机器人的吸附爬行提供了更多的可能性。

Project Introduction

The remora disc and three-legged wheel-based adsorption and crawling structure provides better safety and adaptability for the ROV, which can effectively help the ROV to work stably on the surface of the pipe rack, improve the accuracy of the crack detection results, and provide more possibilities for adsorption and crawling of underwater robots.

